

RAFFLES INSTITUTION

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

1	1				
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BIOLOGY

9648/02

Paper 2 Core Paper

12 September 2011

2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer all questions.

Section B

Answer only **one** question.

At the end of the examination, **hand in your essay SEPARATELY**.
The number of marks is given in brackets [] at the end of each question
or part question.

For Examiner's Use	
SECTION A	
1	/9
2	/13
3	/12
4	/13
5	/12
6	/10
7	/11
SECTION B	
8 or 9	/20
TOTAL	/100

This document consists of **19** printed pages.



Raffles Institution
Internal Examination

Section A

Answer all questions in this section.

*For
Examiner's
Use*

- 1 (a) Describe how enzymes take part in chemical reactions. [4]

(b) Lysozyme is an enzyme found in many animals. Lysozyme catalyses the hydrolysis of glycosidic bonds in certain polysaccharides found in the cell walls of some bacteria. In an investigation of the effects of lysozyme, researchers isolated the enzyme from mice to find out how effective the enzyme was at destroying bacteria.

Four different concentrations of lysozyme were made. Two pathogenic bacteria, *Escherichia coli* and *Staphylococcus aureus*, were incubated in each concentration for three hours at 37°C. At the end of the incubation, the researchers determined the number of bacteria still alive and expressed their results as percentages of the number of bacteria present at the start of the incubation. The results are shown in **Fig. 1.1**.

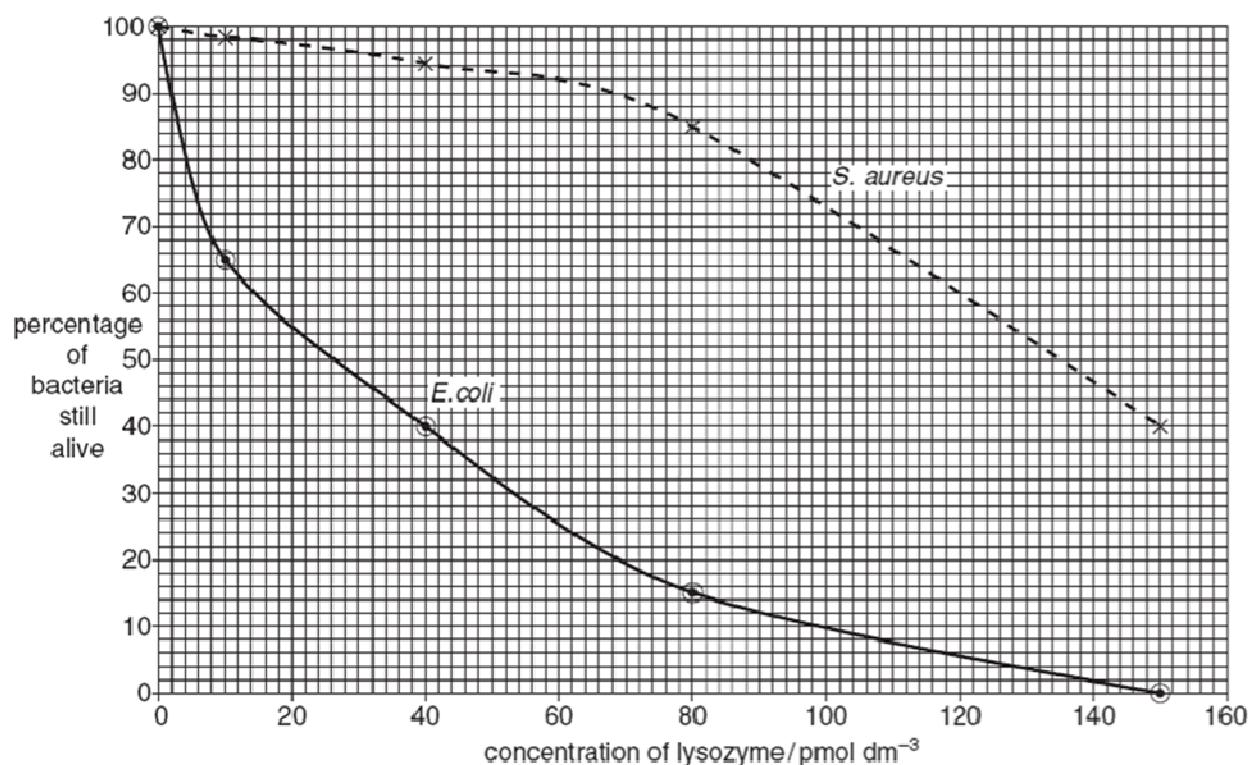


Fig 1.1

- (i) Using the information in the **Fig. 1.1**, describe the effect of the different concentrations of lysozyme on *E. coli* and *S. aureus*. [4]

- (ii) Suggest a possible explanation for the different effects of lysozyme on *E. coli* and *S. aureus*. [1]

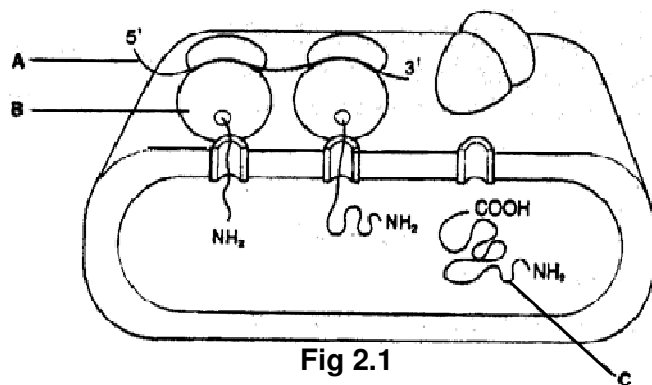
[Total: 9m]

- 2** The main function of a chromosome is to act as a **template** for the synthesis of RNA molecules, since only in this way does the genetic information stored in chromosomes become directly useful to the cell. RNA synthesis is a highly selective process.

- (a)** Explain what is meant by the term **template**. [3]

Fig. 2.1 is a section through the rough endoplasmic reticulum representing the synthesis of proteins.

For
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(b) Describe and explain the role in protein synthesis of

(i) structure **A**; [3]

(ii) structure **B**. [3]

(c) Explain why mutations in the coding region for **C** can still result in identical polypeptide C? [2]

(d) **C** is an insulin receptor. Outline the route taken by protein **C** after it is synthesized in the rough endoplasmic reticulum. [2]

[Total: 13]

3 (a) Fig 3.1 shows the structure of the tryptophan (*trp*) operon of *Escherichia coli*.

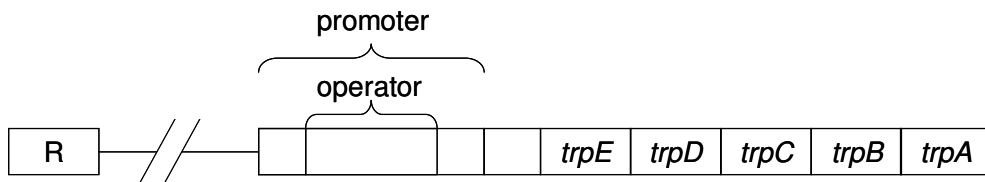


Fig. 3.1

(i) R is a regulatory gene located far away from the *trp* operon but it can still regulate expression of the operon. Why? [2]

- (ii) Explain how the presence of high levels of tryptophan affects expression of the *trp* operon. [3]

- (b) Operons are either inducible or repressible systems. State the difference in the type of metabolic pathways that the two systems are associated with. [1]

- (c) In May 2011, a strain of *E. coli* caused fatalities, mainly in Germany. This strain of *E. coli* differs from other harmless strains (such as those normally found in our gut) in that they possess genes for a poison known as Shiga toxin. It is believed that these genes were acquired by horizontal gene transfer, possibly via the process of transduction.

Explain how transduction could have led to a strain of a previously harmless *E. coli* gaining genes for the production of Shiga toxin. [4]

(d) Viruses such as bacteriophages have been described as being “organisms at the edge of life”. State one characteristic of viruses that may classify them as being

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Use*

(i) living [1]

(ii) non-living [1]

[Total: 12m]

- 4 Over-expression of the HER2 (human epidermal growth factor receptor 2) gene is often associated with the development of breast cancer. One of the mechanisms causing the over-expression is illustrated in **Fig. 4.1**.

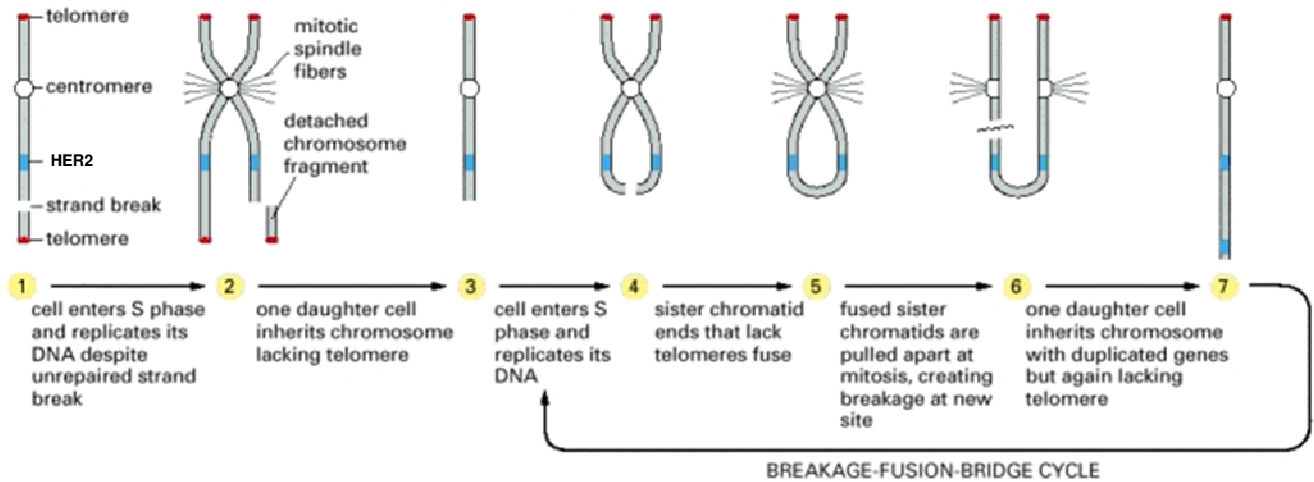


Fig. 4.1

- (a) Name the process that has occurred. [1]

- (b) Explain how the above process can result in over-expression of HER2. [3]

(c) Explain how chromosomal translocation can also result in over-expression of the HER2 gene.
[3]

*For
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Use*

- (d) A similar mechanism exists to increase the expression of rRNA genes in oocytes of the South African clawed toad.

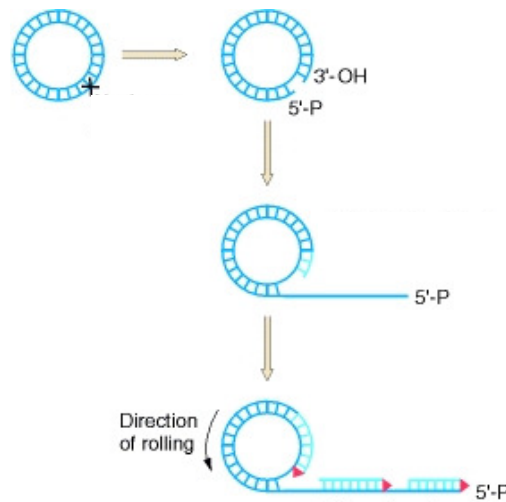


Fig. 4.2

With reference to **Fig. 4.2**, describe the process of replication. [4]

- (e) Explain why such a mechanism is necessary in the oocytes. [2]

[Total: 13m]

- 5 (a) Distinguish between phenotype and genotype. [2]

- (b) Explain how a mutation that results in the substitution of a single base may affect the phenotype of an organism. [3]

- (c) The human disorder *incontinentia pigmenti* is a rare syndrome in which melanin is not retained by melanocytes, causing swirling pigment lines on the skin. Su Lin, a woman with the disorder, marries Hock Seng, a normal man. She gives birth to three normal daughters, two affected daughters, and two normal sons. In addition, she had three miscarriages of deformed male fetuses. (Assume that the syndrome is controlled by a single gene.)

- (i) State the pattern of inheritance of *incontinentia pigmenti*. [2]

- (ii) What is the probability that the next child that Su Lin and Hock Seng have is a healthy and normal female? Explain your answer in the form of a genetic diagram. [5]

For
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Use

Probability of Su Lin and Hock Seng having a healthy normal female child is: _____

[Total: 12m]

6 **Fig. 6.1** is an electron micrograph of a cross section of a neuron.

For
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Use

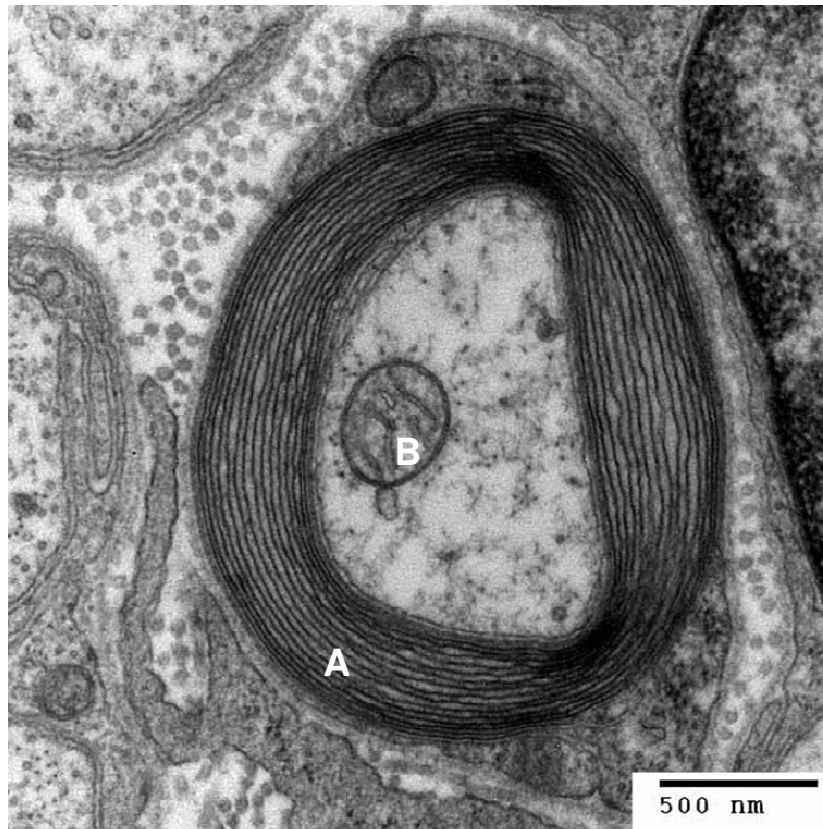
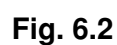


Fig. 6.1

(a) Describe the structure of **A**. [2]

(b) Identify the structure **B** and explain its significance in the transmission of impulse. [3]

-
-
-
-
-



- | Sodium channel | Potassium channel |
|----------------|-------------------|
| | |

[turn over

- 7 (a) **Table 7.1** shows the ancestral and derived traits of five characters (**a** through **e**) for six species (**1** through **6**).

Note: Letters **a** through **e** in the phylogenetic tree show where the specific derived traits appeared. In the table (-) = ancestral state, (+) = derived state.

Derived traits <i>species</i>	a	b	c	d	e
1	-	+	-	+	-
2	+	+	-	+	+
3	-	-	-	+	-
4	+	+	-	+	-
5	-	-	-	-	-
6	-	+	+	+	-

Table 7.1

- (i) For species **1** and **6**, write in the boxes where the two species should be positioned in the phylogenetic tree (**Fig. 7.1**). [2]

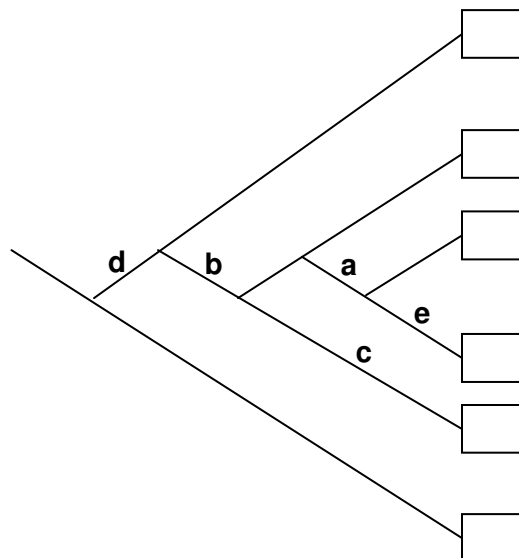


Fig. 7.1

- (ii) On **Fig. 7.1**, mark out a monophyletic group that includes 1 and 6. [1]

(b) Fig. 7.2 shows the possible phylogenetic tree of three mammal groups.

Possible phylogenetic tree of mammals

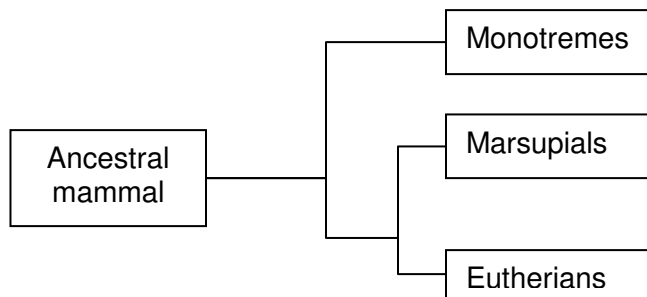


Fig. 7.2

(i) Name one important ancestral character shared by all three mammal groups. [1]

(ii) Name one derived character that both marsupials and eutherians share but is not found among the monotremes. [1]

- (c) Although organisms that are closely related share characteristics because of common descent, distantly related organisms can resemble one another for a different reason: convergent evolution.

Consider marsupial mammals, many of which live in Australia. Marsupials are distinct from another group of mammals – the eutherians – that live elsewhere on earth. Some Australian marsupials have eutherian look-alikes. For instance, a forest-dwelling Australian marsupial called the sugar glider is superficially very similar to flying squirrels, a gliding eutherian that live in North American forests. However, the sugar glider has many other characteristics that make it a marsupial, making it more closely related to kangaroos and other Australian marsupials than to flying squirrels and other eutherians. Once again, our understanding of evolution can explain these observations.

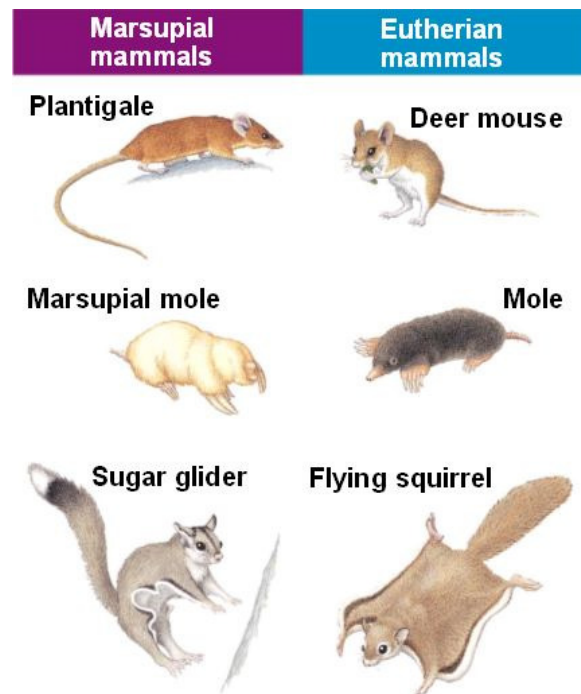


Fig. 7.3

- (i) Explain the term convergent evolution. [1]

- (ii) Explain how the resemblance between various eutherians and Australian marsupial mammals could have evolved. [4]

- (iii) Suggest why there are no native eutherian mammals in Australia. [1]

[Total: 11m]

Section B

Answer **one** question.

Write your answers on the loose sheets of answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a)** Explain how meiosis can lead to variation. [8]
- (b)** Explain why the population is the smallest unit that can evolve. [5]
- (c)** Explain how species are formed with reference to reproductive isolation. [7]
- 9 (a)** Describe the structure and function of centromeres. [7]
- (b)** Outline the differences between prokaryotic transcriptional control of gene expression with the eukaryotic model. [7]
- (c)** Outline the significance of post-translational modifications of eukaryotic proteins. [6]